

复变期末试题

一. 选择题 (5×3分)

1. 下列命题错误的是 D

A. $z\bar{z} = |z|^2$

B. $|z_1 z_2| = |z_1| |z_2|$

C. $|\frac{z_1}{z_2}| = \frac{|z_1|}{|z_2|} (z_2 \neq 0)$

D. $|z_1 + z_2| = |z_1| + |z_2|$

2. 下列命题错误的是 C

A. $e^{z+2\pi i} = e^z$

B. $\ln(z_1 z_2) = \ln z_1 + \ln z_2$

C. $|\cos z| = 1$

D. $\text{Arg}(z_1 z_2) = \text{Arg}(z_1) + \text{Arg}(z_2)$

3. $f(z) = \sin z$, 则下列命题错误的是 C

A. $f(z)$ 在复平面上处处解析

B. $f(z)$ 以 2π 为周期

C. $f(z) = \frac{e^{iz} + e^{-iz}}{2}$

D. $|f(z)|$ 是无界的

4. 下列关于解析函数的描述错误的是 D

A. 在解析的区域内积分与路径无关

B. 解析函数存在任意阶的高阶导数

C. 在点 (x, y) 处解析, 则必在点 (x, y) 处可导

D. 在点 (x, y) 处可导, 则必在点 (x, y) 处解析

5. $\oint_C z e^z dz =$ C, 其中 C 是正向圆周 $|z| = 1$

A. 2π

B. $2i$

C. 0

D. π

$2\pi i \cdot z e^z|_{z=0} = 0$

二. 填空题 (5×3分)

1. 复数 $z = -\sqrt{2} - 2i$ 的极表示为 $4(\cos \frac{5\pi}{4} - i \sin \frac{5\pi}{4})$

2. $\ln(1-i) =$ $\frac{1}{2}\ln 2 - \frac{\pi}{4}i$

3. 积分 $\int_0^1 z \cos z dz =$ $e^{-1} - 1$ $\int_0^1 z \cos z dz = \int_0^1 z d \sin z = z \sin z|_0^1 - \int_0^1 \sin z dz = i \sin 1 + \cos z|_0^1 = i \sin 1 + \cos 1 - 1 = e^{i^2} - 1 = e^{-1} - 1$

4. 设 C 为正向圆周 $|z| = 1$, 则 $\oint_C \bar{z} dz =$ 0

5. 单位脉冲函数, 则 δ -函数的 Fourier 变换 $F(\omega) =$ 1

三. 计算 (6x6分)

1. 当 x, y 等于什么实数时 等式 $\frac{x+1+i(y-3)}{5+3i} = 1+i$ 成立?

$$\begin{aligned} x+1+i(y-3) &= (5+3i)(1+i) \\ &= 5+5i+3i-3 \\ &= 2+8i \end{aligned} \quad \begin{aligned} x+1 &= 2 \\ y-3 &= 8 \end{aligned} \Rightarrow \begin{cases} x=1 \\ y=11 \end{cases}$$

2. 求方程 $z^5+32=0$ 的所有根

$$z = \sqrt[5]{-32} \quad \because -1 = \cos \pi + i \sin \pi$$

$$\therefore z = \sqrt[5]{-32} = \sqrt[5]{32(\cos \pi + i \sin \pi)} = 2(\cos \pi + i \sin \pi)^{\frac{1}{5}} = 2\left(\cos \frac{\pi+2k\pi}{5} + i \sin \frac{\pi+2k\pi}{5}\right)$$

$$k=0, 1, 2, 3, 4$$

$$z_0 = 2(\cos \frac{\pi}{5} + i \sin \frac{\pi}{5}) \quad z_1 = 2(\cos \frac{3\pi}{5} + i \sin \frac{3\pi}{5})$$

$$z_2 = 2(\cos \frac{5\pi}{5} + i \sin \frac{5\pi}{5}) = 2(\cos \pi + i \sin \pi) = -2$$

$$z_3 = 2(\cos \frac{7\pi}{5} + i \sin \frac{7\pi}{5})$$

$$z_4 = 2(\cos \frac{9\pi}{5} + i \sin \frac{9\pi}{5})$$

3. $\int_C z dz$, 其中 C 为从原点到点 $3+4i$ 的直线段.

直线段方程写成 $x=3t, y=4t, 0 \leq t \leq 1$

$$z = 3t + 4ti = (3+4i)t, 0 \leq t \leq 1$$

$$\int_C z dz = \int_0^1 (3+4i)t d(3+4i)t = \int_0^1 (3+4i)^2 t dt = \frac{(3+4i)^2}{2} t^2 \Big|_0^1 = \frac{(3+4i)^2}{2}$$

4. $\oint_C \frac{1}{(z+2)(z-1)} dz$, 其中 $C: |z|=4$ 正向

$$\oint_C \frac{1}{(z+2)(z-1)} dz = \oint_C \left(\frac{\frac{1}{3}}{z-1} - \frac{\frac{1}{3}}{z+2} \right) dz = -\frac{1}{3}$$

$$= \oint_C \frac{\frac{1}{3}}{z-1} dz + \oint_C \frac{\frac{1}{3}}{z+2} dz = 2\pi i \cdot \frac{1}{3} \Big|_{z=1} + 2\pi i \cdot \frac{1}{3} \Big|_{z=-2} = \frac{2}{3}\pi i - \frac{2}{3}\pi i = 0$$

5. $\oint_C \frac{\cos z}{z^2} dz$, 其中 $C: |z|=2$ 正向

$$\oint_C \frac{\cos z}{z^2} dz = 2\pi i (\cos z)' \Big|_{z=0} = 0$$

6. 求 $f(t) = \cos kt$ 的 Laplace 变换式.

$$\text{由于 } f(0)=1, f'(0)=0, f''(t)=-k^2 \cos kt$$

$$\text{微分性质 } \mathcal{L}(-k^2 \cos kt) = \mathcal{L}[f''(t)] = s^2 \mathcal{L}[f(t)] - s \mathcal{L}[f'(0)] - f'(0)$$

$$-k^2 \mathcal{L}[\cos kt](s) = s^2 \mathcal{L}[\cos kt](s) - s$$

$$\text{得 } \mathcal{L}(\cos kt) = \frac{s}{s^2+k^2} \quad (\operatorname{Re}(s) > 0)$$

$$(\operatorname{Re}(s) > 0)$$

10(10分) 设 $f(z) = my^3 + nx^2y + i(x^3 + lxy^2)$ 为解析函数. 试确定 l, m, n 的值. 并求 $f'(z)$
 $u = my^3 + nx^2y \quad v = x^3 + lxy^2$

$$\therefore f(z) \text{ 为 } \frac{\partial u}{\partial x} = 2nx, \frac{\partial u}{\partial y} = 3my^2 + nx^2$$

$$\frac{\partial v}{\partial x} = 3x^2 + ly^2 \quad \frac{\partial v}{\partial y} = 2lxy$$

$\therefore f(z)$ 为解析函数 即满足柯西-黎曼方程 (C-R 方程)

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y} \quad \frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}$$

$$2nx = 2lxy$$

$$3my^2 + nx^2 = -3x^2 - ly^2 \Rightarrow \begin{cases} l = -3 \\ n = -3 \\ m = 1 \end{cases}$$

(12分)

五. 证明 $u(x, y) = 2y^3 - 6x^2y$ 为调和函数. 并求其共轭调和函数 $v(x, y)$ 及解析函数 $f(z)$

$$\frac{\partial u}{\partial x} = -12xy \quad \frac{\partial^2 u}{\partial x^2} = -12y \quad \frac{\partial u}{\partial y} = 6y^2 - 6x^2 \quad \frac{\partial^2 u}{\partial y^2} = 12y$$

$$\therefore \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0 \quad \therefore u(x, y) \text{ 是调和函数.}$$

设 $v(x, y)$ 是其共轭调和函数 $\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y} = -12y \quad \frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x} = 6y^2 - 6x^2$

$$\frac{\partial v}{\partial y} = -12xy \quad v = \int -12xy dy = -6xy^2 + g(x)$$

$$\frac{\partial v}{\partial x} = -12xy = -\frac{\partial v}{\partial x} \quad \frac{\partial v}{\partial x} = -6y^2 + g'(x) = 6x^2 - 6y^2 \quad \therefore g'(x) = 6x^2$$

$$g(x) = \int 6x^2 dx = 2x^3 + C \quad \therefore v = -6xy^2 + 2x^3 + C$$

$$\therefore f(z) = 2y^3 - 6x^2y + (-6xy^2 + 2x^3 + C)i = 2iz^3 + ic$$

六. (12分) 求指数衰减函数 $f(t) = \begin{cases} 0 & t < 0 \\ e^{-\beta t} & t \geq 0 \end{cases} (\beta > 0)$ 的 Fourier 变换. 并由此用 Fourier 变换解以下微分方程和解.

$$x'(t) + x(t) = \delta(t), \quad (-\infty < t < +\infty)$$

$$F(\omega) = \int_{-\infty}^{+\infty} f(t)e^{-i\omega t} dt = \int_0^{+\infty} e^{-\beta t} e^{-i\omega t} dt = \int_0^{+\infty} e^{-(\beta + i\omega)t} dt = \frac{1}{\beta + i\omega} = \frac{\beta - i\omega}{\beta^2 + \omega^2}$$

设 $x(t)$ 的 Fourier 变换为 $X(\omega)$. 将方程两边同时取 Fourier 变换

$$i\omega X(\omega) + X(\omega) = 1$$

$$\begin{aligned} X(\omega)(i\omega + 1) &= 1 \\ X(\omega) &= \frac{1}{i\omega + 1} \end{aligned}$$

$$\text{取 Fourier 逆变换得方程解为 } x(t) = \begin{cases} 0, & t < 0 \\ e^{-t}, & t \geq 0 \end{cases}$$